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Smooth Vector Fields

1.1 Vector Fields on Manifold

Definition 1.1.1

If M is a smooth manifold with or without boundary, a ***vector field on M*** is a section of the map $\pi : TM \to M$. More concretely, a vector field is a continuous map $X : M \to TM$, usually written $p \mapsto X_p$, with the property that

$$\pi \circ X = \mathrm{Id}_M, \tag{1.1}$$

or equivalently, $X_p \in T_pM$ for each $p \in M$.

Remark.

- For a "vector field" that may be not contiuous, we call it ***rough vector field on M***.
- Under the canonical smooth structure of M and TM, we can define the smoothness of X.

Definition 1.1.2 ▸ component functions of a vector field

Suppose M is a smooth n-manifold (with or without boundary). If $X : M \to TM$ is a rough vector field and $(U, (x^i))$ is any smooth coordinate chart for M, we can write the value of X at any point $p \in U$ in terms of the coordinate basis vectors:

$$X_p = X^i(p) \frac{\partial}{\partial x^i}\bigg|_p . \tag{1.2}$$

This defines n functions $X^i : U \to \mathbb{R}$, called the ***component functions of X*** in the given chart.

Proposition 1.1.3 ▸ Smoothness Criterion for Vector Fields

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. If $(U, (x^i))$ is any smooth coordinate chart on M, then the restriction of X to U is smooth if and only if its component functions with respect to this chart are smooth.

Proposition 1.1.4

Let M be a smooth manifold with or without boundary, and let $X : M \to TM$ be a rough vector field. The following are equivalent:

(a) X is smooth.

(b) For every $f \in C^\infty(M)$, the function Xf is smooth on M.

(c) For every open subset $U \subseteq M$ and every $f \in C^\infty(U)$, the function Xf is smooth on U.

Definition 1.1.5 ▸ Vector Field as a Derivation

A smooth vector field $X \in \mathfrak{X}(M)$ defines a map

$$\begin{aligned} X : C^\infty(M) &\to C^\infty(M) \\ f &\mapsto Xf \end{aligned}$$

which is linear over $\mathbb{R}$ and following the product rule for vector fields

$$X(fg) = fXg + gXf$$

We call this map a ***Derivation on*** $C^\infty(M)$.

Proposition 1.1.6

Let M be a smooth manifold with or without boundary. A map $D : C^\infty(M) \to C^\infty(M)$ is a derivation if and only if it is of the form $Df = Xf$ for some smooth vector field $X \in \mathfrak{X}(M)$.

1.2 Lie Brackets

Definition 1.2.1 ▶ Lie Barckets

The operator $[X, Y] : C^\infty(M) \to C^\infty(M)$, called the ***Lie bracket of*** X ***and*** Y, defined by

$$[X, Y]f = XYf - YXf.$$

is a vector field.